E XP.NO -8	MINI PROJECT-EV Market & Charging Intelligence

AIM:

To clean, analyze, and visualize electric vehicle and charging infrastructure data using Python, Power BI, and Tableau to identify EV market trends, vehicle performance, battery characteristics, charging capabilities, and charging infrastructure patterns.

DOMAIN:

Electric Vehicle Analytics / Automotive Analytics / Business Intelligence.

PROBLEM STATEMENT:

The growth of electric vehicles has created a need to understand vehicle performance, battery capacity, driving range, charging capabilities, and charging infrastructure. This project analyzes EV and charging-station data using Python, Power BI, and Tableau to generate meaningful insights and support data-driven comparison.

DATASET DESCRIPTION:

The project uses two public datasets.

1. Electric Vehicle Specifications Dataset 2025

The vehicle dataset contains 478 EV records and 22 fields covering information such as:

- EV manufacturer
- EV model
- Battery capacity
- Energy efficiency
- Driving range
- Fast-charging power
- Top speed
- Torque
- Acceleration
- Towing capacity
- Cargo volume
- Number of seats
- Vehicle dimensions
- Drivetrain
- Segment
- Body type

2. Global EV Charging Stations 2025

The charging infrastructure dataset contains 242,417 charging sites across 121 countries with 11 fields, including:

- Charging station ID
- Station name
- City
- State/Province
- Country code
- Latitude
- Longitude
- Number of charging ports
- Maximum charging power
- Power class
- Fast-DC availability

The two datasets represent different entities and therefore were not directly merged at row level. Instead, they were analyzed separately and connected conceptually through the Power BI and Tableau dashboards.

PROCEDURE:

1. Loaded the EV and charging infrastructure datasets using Python and Pandas.
2. Inspected dataset structure, dimensions, data types, missing values, and duplicate records.
3. Cleaned and standardized text and numerical fields.
4. Removed duplicate records and invalid physical measurements.
5. Validated charging station geographic coordinates.
6. Standardized country codes and fast-DC indicators.
7. Created derived EV performance indicators.
8. Created charging infrastructure indicators and power categories.
9. Performed exploratory data analysis using Pandas, Matplotlib, and Seaborn.
10. Analyzed relationships between battery capacity, driving range, efficiency, and charging power.
11. Analyzed charging infrastructure by country, power category, and fast-DC availability.
12. Built Linear Regression and Random Forest models for EV driving-range prediction.
13. Compared model performance using MAE, RMSE, and R^2.
14. Created static and interactive Python visualizations.
15. Exported cleaned datasets and summary tables.
16. Created an interactive EV Market Intelligence Dashboard in Power BI.
17. Created an interactive EV & Charging Infrastructure Explorer in Tableau.

KEY INSIGHTS:

The analysis will investigate:

- EV manufacturers with the highest number of models.
- Distribution of EV battery capacities.
- Distribution of EV driving ranges.
- Relationship between battery capacity and driving range.
- Relationship between energy efficiency and driving range.
- Relationship between battery capacity and fast-charging power.
- Average driving range across vehicle body types.
- EV models with the longest driving ranges.
- Correlations between major EV technical specifications.
- Countries with the largest charging infrastructure.
- Countries with the highest number of charging ports.
- Distribution of charging station power.
- Charging stations by power category.
- Fast-DC charging availability.
- Fast-DC share across countries.
- Geographic distribution of charging stations.
- Prediction of EV driving range using machine-learning models.

PYTHON ANALYSIS

Python was used for data cleaning, processing, exploratory data analysis, feature engineering, and visualization.

The main libraries used were:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- Scikit-learn

DATA CLEANING

The datasets were loaded using Pandas.

The following operations were performed:

- Checked dataset structure and dimensions.
- Examined data types.
- Identified missing values.
- Identified duplicate records.
- Standardized text values by removing unnecessary spaces.
- Converted numerical fields into appropriate numeric data types.
- Removed duplicate EV records.
- Checked numerical fields for zero and negative values.
- Converted invalid negative physical measurements to missing values.
- Standardized charging station country codes.
- Converted fast-DC values into Boolean format.
- Removed duplicate charging station IDs.
- Validated latitude and longitude values.

The vehicle dataset contained 478 records after cleaning, while the charging dataset contained 242,417 charging sites after cleaning.

DATA PROCESSING / FEATURE ENGINEERING

1. Range per kWh

$$\text{Range per kWh} = \text{Driving Range} / \text{Battery Capacity}$$

This indicator helps compare the amount of driving range obtained from each unit of battery capacity.

2. Charging Power per kWh

$$\text{Charging Power per kWh} = \text{Fast Charging Power} / \text{Battery Capacity}$$

This provides an analytical comparison between battery size and fast-charging capability.

3. Estimated kWh per 100 km

$$\text{kWh per 100 km} = \text{Battery Capacity} / \text{Driving Range} \times 100$$

This provides an estimated energy-use indicator based on battery capacity and rated driving range.

4. Range Efficiency Index

$$\text{Range Efficiency Index} = \text{Driving Range} / \text{Energy Efficiency}$$

This provides an additional analytical indicator for comparing EV performance characteristics.

5. Range Category

EVs were grouped into four range categories:

- Short
- Medium
- Long
- Very Long

These categories were used to support dashboard filtering and comparison.

6. Charging Power per Port

$$\text{Power per Port} = \text{Charging Power} / \text{Number of Ports}$$

This provides an analytical measure of charging power relative to the number of available ports.

7. Charging Power Category

Charging stations were classified according to their maximum power into:

- Very Low
- AC Standard
- Fast
- High Power
- Ultra Fast
- Extreme

This classification was used to analyze the distribution of charging infrastructure

EXPLORATORY DATA ANALYSIS

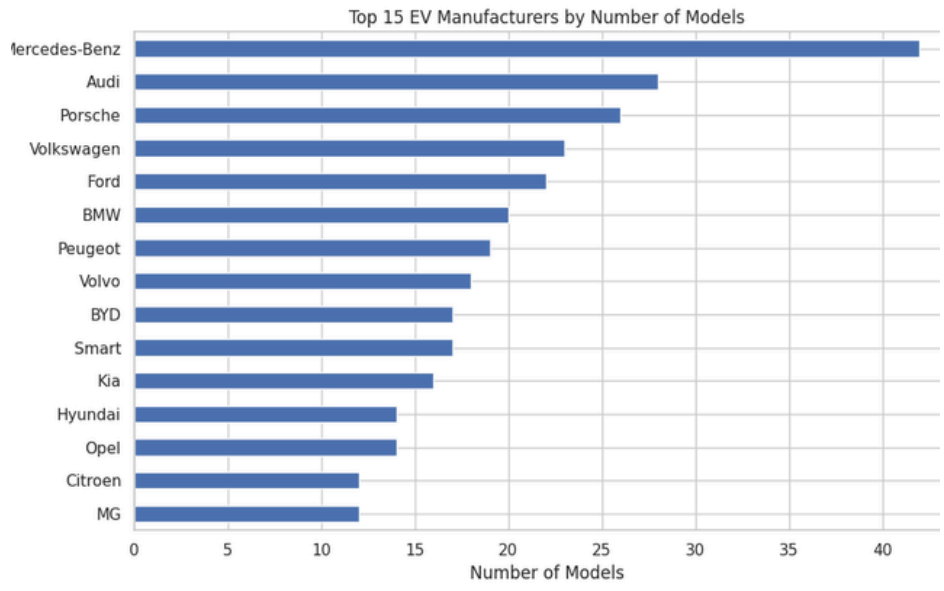
Python will be used to analyze:

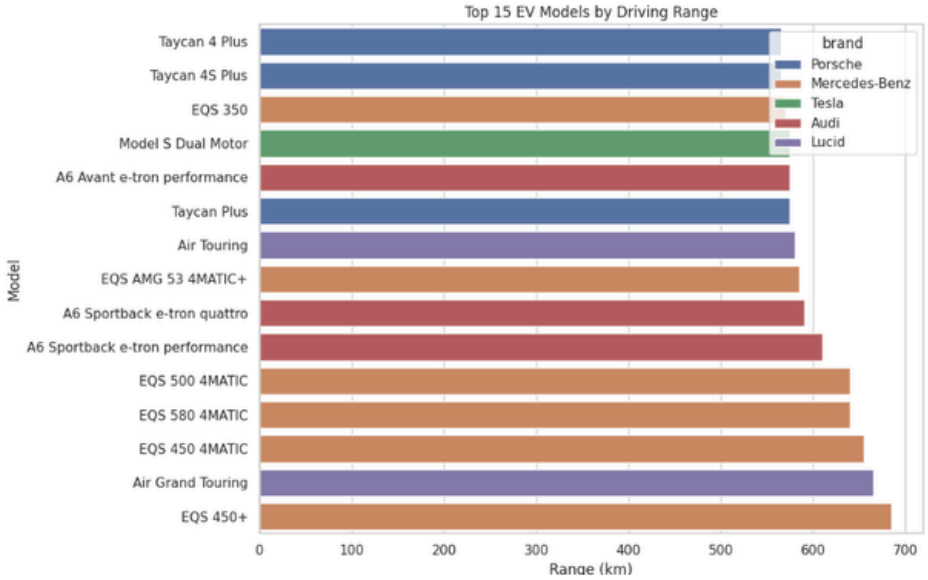
- EV models by manufacturer
- Battery capacity distribution
- Driving range distribution
- Battery capacity versus driving range

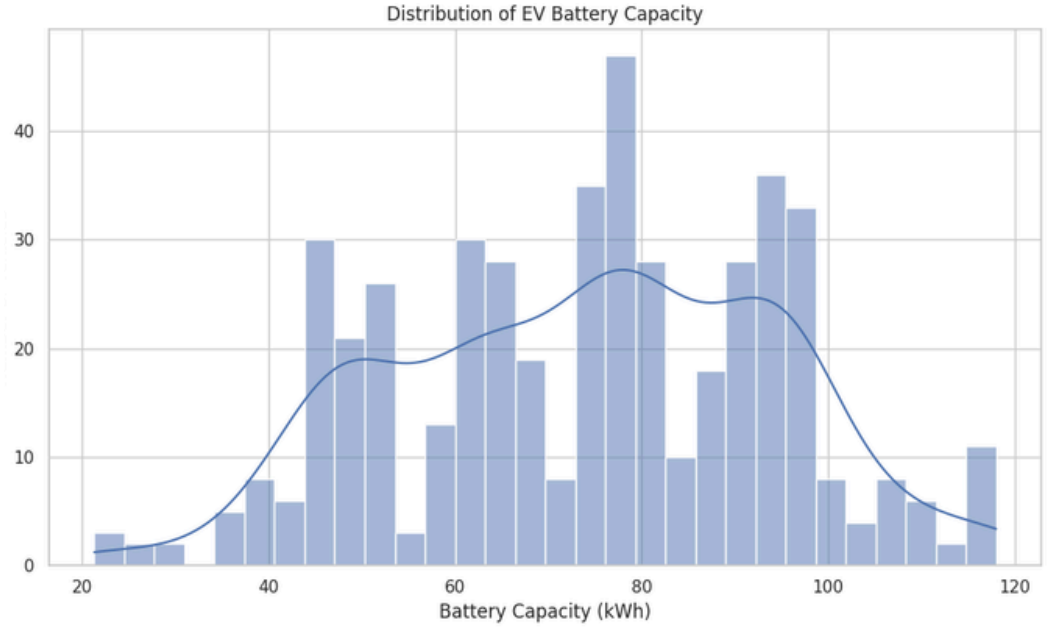
- Energy efficiency versus driving ranger
- Battery capacity versus fast-charging powe
- Average range by body type
- Top EV models by driving range
- Correlation between EV technical features
- Charging infrastructure by country
- Charging ports by country
- Fast-DC versus other charging stations
- Charging power distribution
- Charging stations by power category
- Fast-DC share by country
- Geographic distribution of charging stations
- Actual versus predicted EV driving range

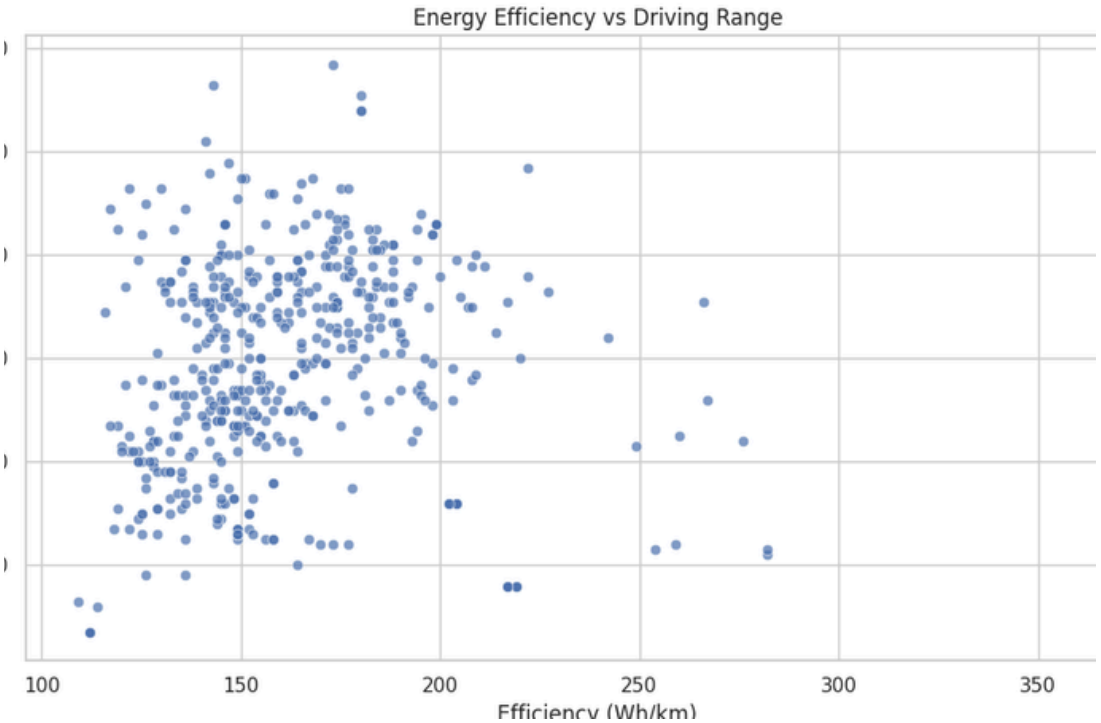
PYTHON VISUALIZATIONS

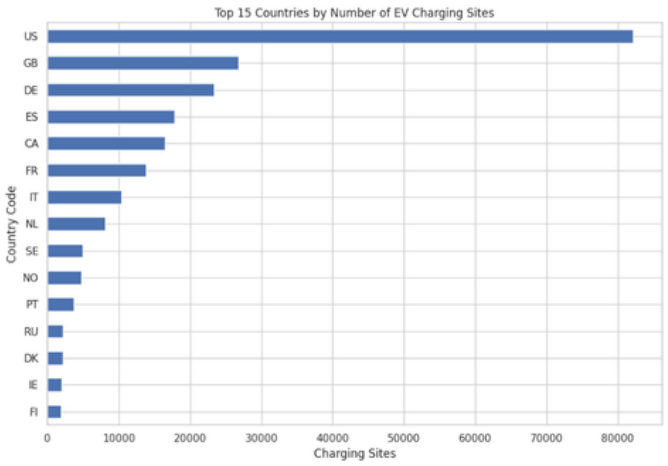
Python visualizations were created using Matplotlib and Seaborn to analyze EV manufacturers, battery capacity, driving range, energy efficiency, and charging infrastructure. Additional visualizations examined charging station power and the relationship between actual and predicted EV driving range using machine learning. The results were used for the Power BI and Tableau dashboards.

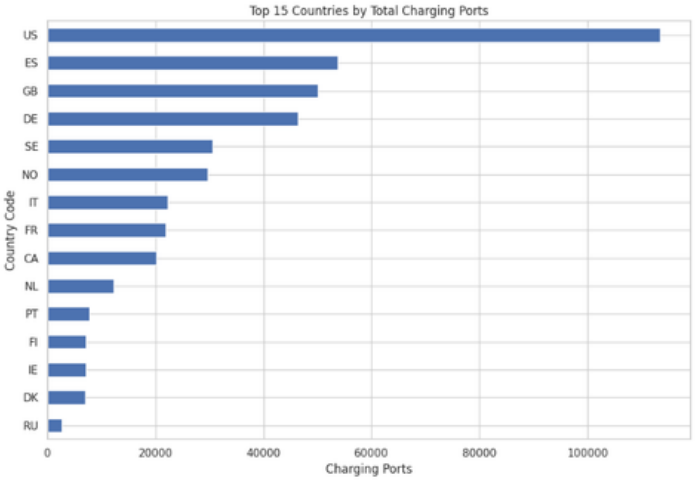


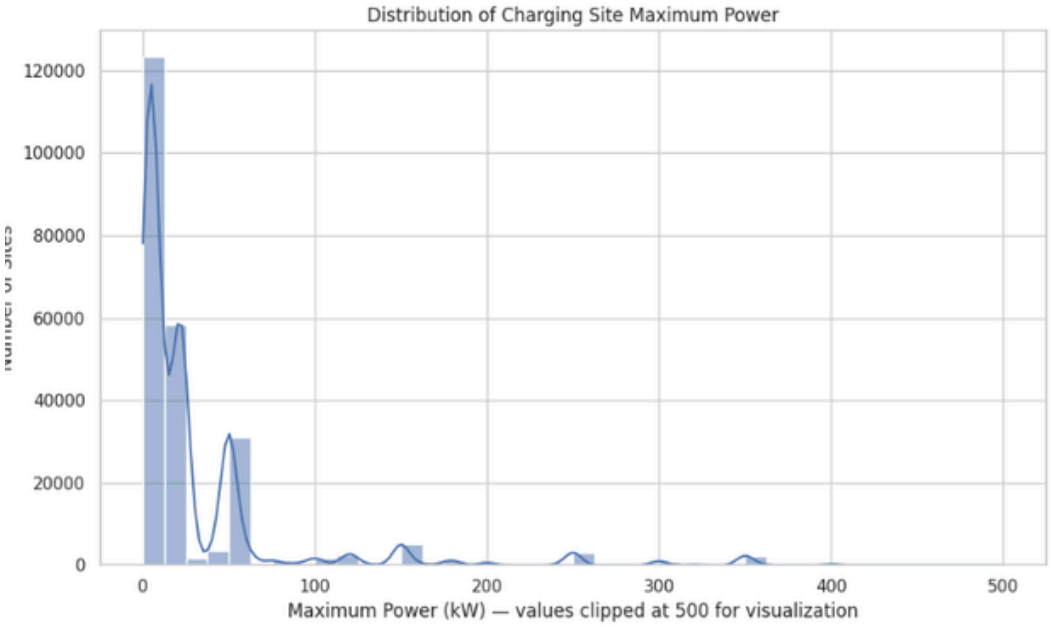

Top 15 EV Models by Driving Range
Taycan 4 Plus
Taycan 4S Plus
EQS 350
Model S Dual Motor
A6 Avant e-tron performance
Taycan Plus
Air Touring
EQS AMG 53 4MATIC+
A6 Sportback e-tron quattro
A6 Sportback e-tron performance
EQS 500 4MATIC
EQS 580 4MATIC
EQS 450 4MATIC
Air Grand Touring
EQS 450+
Model
brand
Porsche
Mercedes-Benz
Tesla
Audi
Lucid
0
100
200
300
400
500
600
700
Range (km)

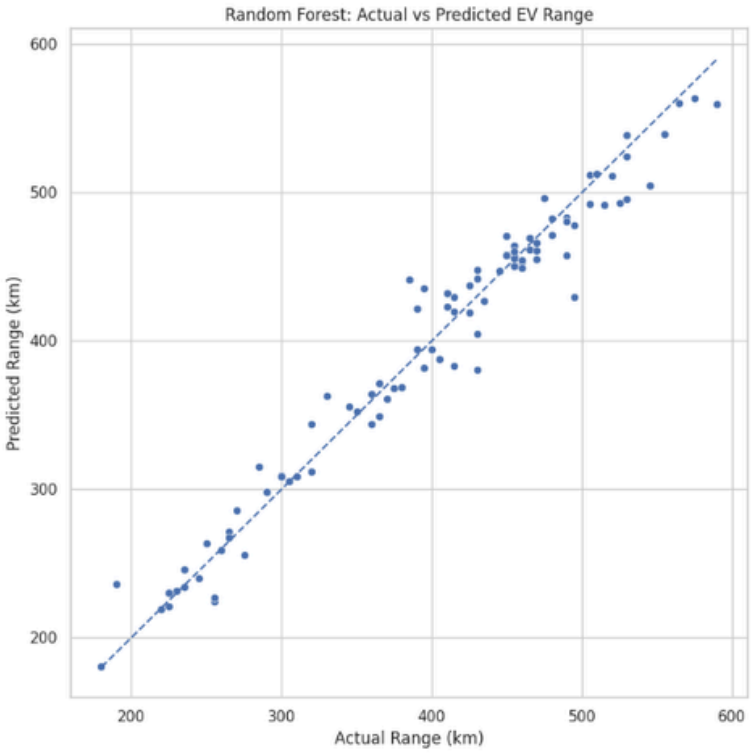

Distribution of EV Battery Capacity
0
10
20
30
40
20
40
60
80
100
120
Battery Capacity (kWh)


Energy Efficiency vs Driving Range
100
150
200
250
300
350
Efficiency (Wh/km)


Top 15 Countries by Number of EV Charging Sites
US
GB
DE
ES
CA
FR
IT
NL
SE
NO
PT
RU
DK
IE
FI
Country Code
0
10000
20000
30000
40000
50000
60000
70000
80000
Charging Sites

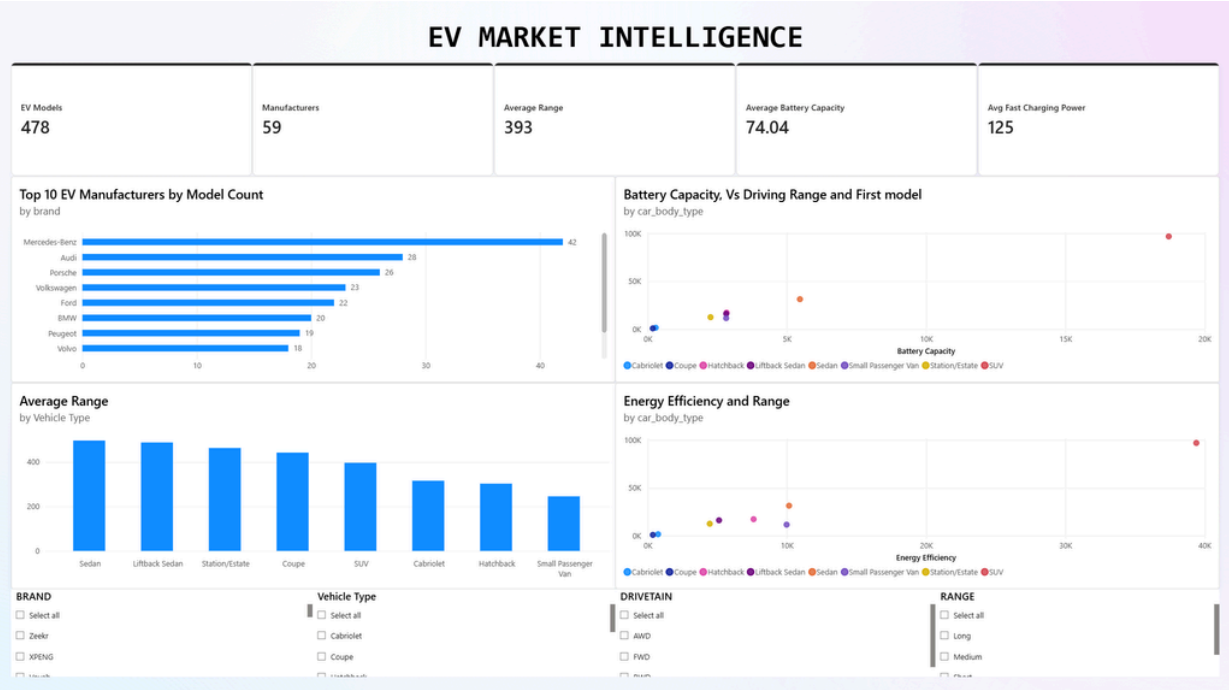

Top 15 Countries by Total Charging Ports
US
ES
GB
DE
SE
NO
IT
FR
CA
NL
PT
FI
IE
DK
RU
Country Code
0
20000
40000
60000
80000
100000
Charging Ports


Distribution of Charging Site Maximum Power
120000
100000
80000
60000
40000
20000
0
0
100
200
300
400
500
Maximum Power (kW) — values clipped at 500 for visualization


Random Forest: Actual vs Predicted EV Range
600
500
400
300
200
Predicted Range (km)
200
300
400
500
600
Actual Range (km)

Power BI Output

The cleaned datasets were imported into Power BI to create interactive EV Market Intelligence and Global EV Charging Infrastructure dashboards. KPI cards and visualizations are used to analyze EV models, manufacturers, battery capacity, driving range, energy efficiency, vehicle types, charging power, charging stations, charging ports, fast DC charging, and the geographical distribution of charging infrastructure. Interactive slicers are also provided for vehicle and charging-related categories to support detailed analysis and comparison.



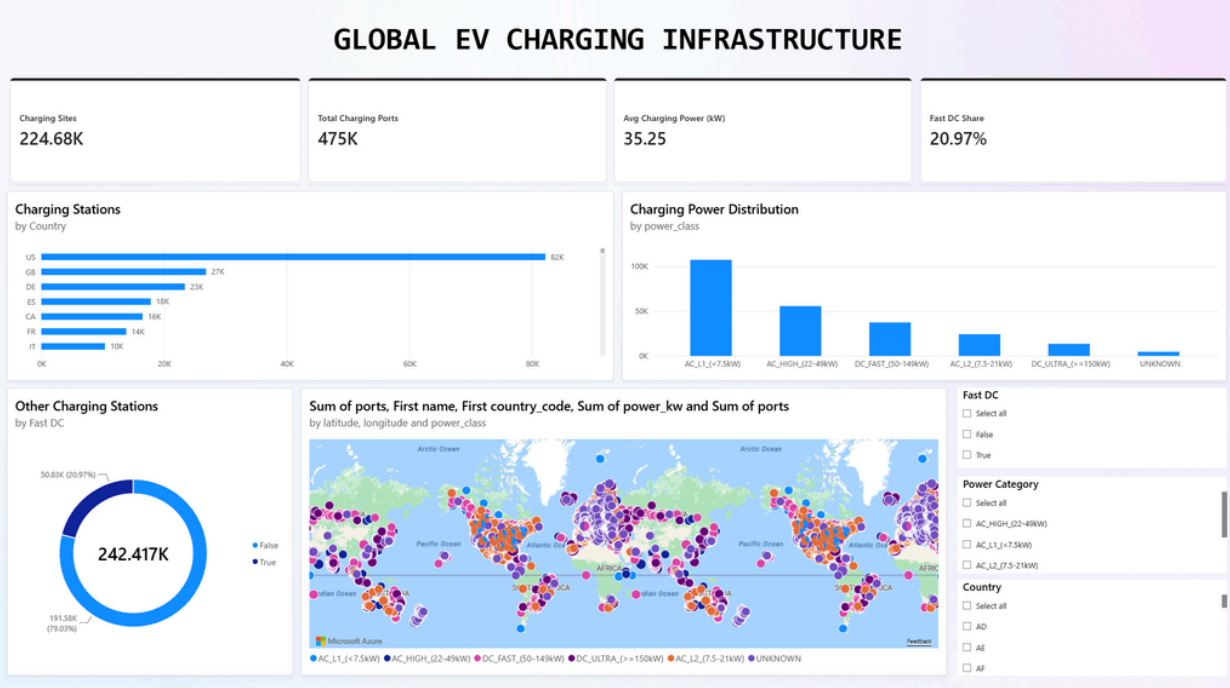
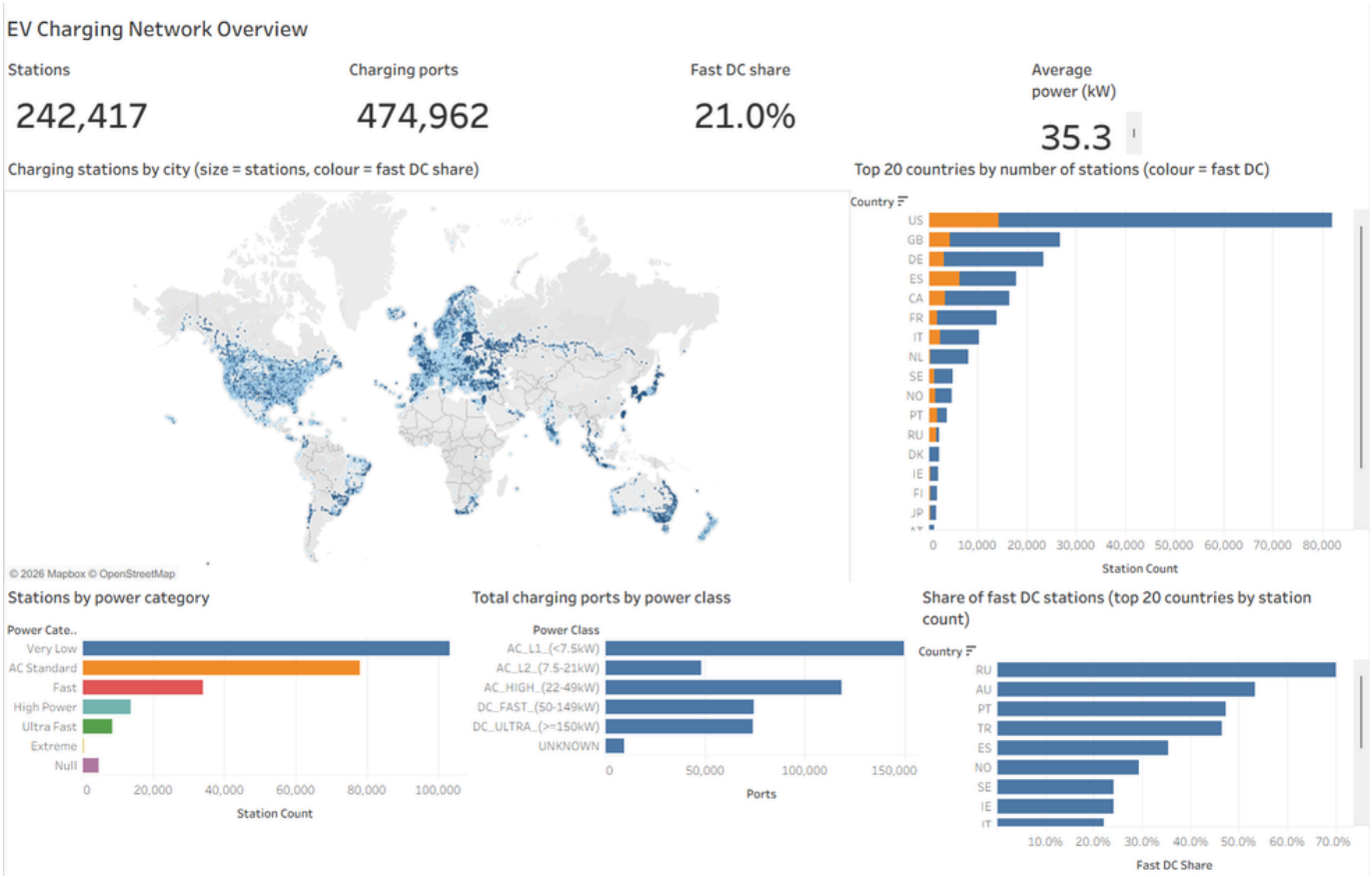


Tableau Output

The cleaned datasets were also used in Tableau to create an interactive EV & Charging Infrastructure Explorer. A geographic map, charging infrastructure analysis, manufacturer analysis, and comparative visualizations are used to explore charging infrastructure, EV performance, battery characteristics, charging power, and relationships between different vehicle specifications. The dashboard includes KPI cards, country-wise charging station analysis, charging ports by power class, fast DC charging share, and charging station categories to provide an interactive overview of the EV market and charging network.



RESULT:

The project successfully transforms raw electric vehicle and charging infrastructure data into a cleaned analytical dataset using Python and presents the resulting insights through interactive Power BI and Tableau dashboards. The project provides a comprehensive view of EV market trends, vehicle performance, battery characteristics, charging capabilities, and the geographical distribution of charging infrastructure. The visualizations help in understanding EV manufacturers, driving range, energy efficiency, charging station distribution, charging power, and fast DC charging patterns.